Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/01/2008
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Analysis of a Novel Sensory Mechanism in Root Phototropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
PI Address 1:	College of Arts & Sciences, 105 Foust Building		
PI Address 2:	PO Box 26170		
PI Web Page:			
City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	07/31/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS NOTE: Received extension to 07/31/2009 per A. Chu (6/08) NOTE: Received extension to 07/31/2008 per A. Chu (5/07) Note anniversary date changed to 8/1; start date remains the same (per A. Chu 4-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University) Correll, Melanie (University of Florida)		
Grant/Contract No.:	NCC2-1200		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of Arabidopsis. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. The phototropic response in roots is much weaker than the graviresponse, which competes with and often masks the phototropic response. It was through the use of mutant plants with a weakened gravitropism that we were able to identify the activity of the red-light-dependent phototropic system. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. In this project, we utlized the European Modular Cultivation System (EMCS) on the ISS. The experiment was performed on the ISS Oct.-Dec. 2006. The last samples were returned to earth on STS-120 in Nov. 2007. This information is important for growing plants on long-term space missions as part of life support systems and advances the current exploration agenda at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	TROPI was performed in the EMCS on the ISS. We expected all samples to be returned in late 2006, however, these samples were not returned until STS-120 in November 2007. Thus, we have been performing analysis of samples in terms of retrieval of data from the video tapes and performing microarray studies.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kiss JZ, Kumar P, Correll MJ, Edelmann RE. "The use of microgravity to study phototropism–the TROPI experiments on the ISS." International Committee on Space Research (COSPAR) Meeting, Montreal, Canada, July 10-13, 2008. International Committee on Space Research (COSPAR) Meeting, Montreal, Canada, July 10-13, 2008. COSPAR Abstract Book. , Jul-2008
Abstracts for Journals and Proceedings	Correll MJ, Stimpson AJ, Pereira RS, Kiss JZ. "The effect of microgravity on the gene expression of Arabidopsis." International Committee on Space Research (COSPAR) Meeting, Montreal, Canada, July 10-13, 2008. International Committee on Space Research (COSPAR) Meeting, Montreal, Canada, July 10-13, 2008. COSPAR Abstract Book. , Jul-2008
Abstracts for Journals and Proceedings	Stimpson AJ, Pereira RS, Kiss JZ, Correll MJ. "Gene expression from plants grown on the International Space Station." International Committee on Space Research (COSPAR) Meeting, Montreal, Canada, July 10-13, 2008. International Committee on Space Research (COSPAR) Meeting, Montreal, Canada, July 10-13, 2008. COSPAR Abstract Book. , Jul-2008
Abstracts for Journals and Proceedings	McCoshum S, Kumar P, Kiss JZ. "Suppressor of phytochrome A (SPA1): an element in phototropic responses." American Society of Plant Biologists, Midwestern Section 2008 Meeting, Ames, IA, March 29-30, 2008. American Society of Plant Biologists, Midwestern Section 2008 Meeting, Ames, IA, March 29-30, 2008. Abstract Book, p. 29. , Mar-2008
Abstracts for Journals and Proceedings	Kumar P, Butler B, Correll MJ, Edelmann RE, Kiss JZ. "Phototropism of Arabidopsis thaliana in microgravity." American Society of Plant Biologists, Midwestern Section 2008 Meeting, Ames, IA, March 29-30, 2008. American Society of Plant Biologists, Midwestern Section 2008 Meeting, Ames, IA, March 29-30, 2008. Abstract Book, p. 16. , Mar-2008
Abstracts for Journals and Proceedings	Kumar N, Kiss JZ. "Characterization of ARG1 in Arabidopsis gravity perceiving cells using immunocytochemistry." American Society of Plant Biologists, Midwestern Section 2008 Meeting, Ames, IA, March 29-30, 2008. American Society of Plant Biologists, Midwestern Section 2008 Meeting, Ames, IA, March 29-30, 2008. Abstract Book, p. 9. , Mar-2008
Abstracts for Journals and Proceedings	Butler BA, Kumar P, Kiss JZ. "Study of phototropism in Arabidopsis thaliana in microgravity on the ISS." American Society for Gravitational and Space Biology 23rd annual meeting, Moffett Field, CA, October 25-28, 2007. Gravitational and Space Biology 2007 Oct;21(1):4. , Oct-2007
Abstracts for Journals and Proceedings	Molas ML, Kiss JZ. "PKS1 is involved in red-light-based positive phototropism in Arabidopsis roots." American Society for Gravitational and Space Biology 23rd annual meeting, Moffett Field, CA, October 25-28, 2007. Gravitational and Space Biology 2007 Oct;21(1):8. , Oct-2007
Abstracts for Journals and Proceedings	Reboulet JC, Kumar P, Kiss JZ. "Role of the actin related protein (ARP) 2/3 complex in gravitropism." American Society for Gravitational and Space Biology 23rd annual meeting, Moffett Field, CA, October 25-28, 2007. Gravitational and Space Biology 2007 Oct;21(1):9. , Oct-2007
Abstracts for Journals and Proceedings	Kumar P, Montgomery CE, Kiss JZ. "The role of phytochrome C in tropisms." American Society for Gravitational and Space Biology 23rd annual meeting, Moffett Field, CA, October 25-28, 2007. Gravitational and Space Biology 2007 Oct;21(1):33. , Oct-2007
Articles in Peer-reviewed Journals	Palmieri M, Schwind MA, Stevens MH, Edelmann RE, Kiss JZ. "Effects of the myosin ATPase inhibitor 2,3-butanedione monoxime on amyloplast kinetics and gravitropism of Arabidopsis hypocotyls." Physiologia Plantarum 2007 Aug;130(4):613-26. http://dx.doi.org/10.1111/j.1399-3054.2007.00926.x , Aug-2007

Articles in Peer-reviewed Journals	Molas ML, Kiss JZ. "PKS1 plays a role in red-light-based positive phototropism in roots." Plant Cell Environ. 2008 Jun;31(6):842-9. Epub 2008 Feb 5. PMID: 18266898 , Jun-2008
Articles in Peer-reviewed Journals	Molas ML, Kiss JZ. "The effect of column purification on cDNA indirect labelling for microarrays." Plant Methods. 2007 Jun 27;3:9. PMID: 17597522 http://dx.doi.org/10.1186/1746-4811-3-9 , Jun-2007
Articles in Peer-reviewed Journals	Kumar P, Montgomery CE, Kiss JZ. "The role of phytochrome C in gravitropism and phototropism in Arabidopsis thaliana." Functional Plant Biology 2008;35(4):298-305. http://dx.doi.org/10.1071/FP08013 , Jun-2008
Articles in Peer-reviewed Journals	Kumar NS, Stevens MHH, Kiss JZ. "Plastid movement in statocytes of the arg1 (altered response to gravity) mutant." American Journal of Botany 2008 Feb;95(2):177-84. http://www.amjbot.org/cgi/content/abstract/95/2/177 , Feb-2008
Journal/Magazine covers	Palmieri M, Schwind MA, Stevens MH, Edelmann RE, Kiss JZ. "Cover in journal Physiologia Plantarum for article, 'Effects of the myosin ATPase inhibitor 2,3-butanedione monoxime on amyloplast kinetics and gravitropism of Arabidopsis hypocotyls.' " Physiologia Plantarum 2007 Aug;140(4):613-26. http://dx.doi.org/10.1111/j.1399-3054.2007.00926.x , , Aug-2007
Journal/Magazine covers	Molas ML, Kiss JZ. "Cover in journal Plant, Cell, and Environment for article, 'PKS1 plays a role in red-light-based positive phototropism in roots.' " Plant Cell Environ. 2008 Jun;31(6):842-9. Epub 2008 Feb 5. PMID: 18266898 , Jun-2008